SUNKIREDDY BARATH

sunkireddybarath07@gmail.com | 8438784178 | Chennai, Tamil Nadu | linkedin/Barath | github/barath80 | Portfolio

EDUCATION

Chennai Institute of Technology - B.E. Computer Science and Engineering *2024 - 2028*

- Cumulative Grade Point Average (CGPA): 8.78 / 10

PROFESSIONAL EXPERIENCE

Bold Analytics *May 2026 - Jul 2026*
Data Analyst Intern *Maharashtra, India*

- Normalized and deduplicated 600K+ government records into a 3NF PostgreSQL schema, exposed via a secure, scalable REST API.
- Built a 12-year financial model covering 100% of Nifty 100 companies, combining feature engineering with Power BI dashboards.

L2M Labs *Jan 2026 - Mar 2026*
Software Engineer Intern *Chennai, India*

- Designed a concurrency-safe registration backend with transaction locking and row-level checks across 10+ academic departments.
- Built REST API integrations and department-wise data management features supporting real-time, high-volume student registration.

KaizenSpark Tech *Jun 2025 - Jul 2025*
Full Stack Development Intern *Chennai, India*

- Built responsive, production-grade frontend interfaces and successfully launched the company's official website live for public use.
- Improved deployment reliability through Git-based workflows, structured debugging, and staged release management across sprints.

PROJECTS

UrbanTrace AI - ANPR Tracking and City Traffic Analytics | *YOLO11m, ByteTrack, ONNX, OpenCV, FastAPI, PostGIS*

- Built a YOLO11m, ByteTrack and CCT-OCR pipeline on ONNX Runtime reaching 92.2% character accuracy at 18.2 fps per camera.
- Reconstructed cross-camera vehicle trajectories with PostGIS and GraphHopper routing; dedupe cut 110 tracker IDs to 28 vehicles.

AgriSmart - AI/ML-Powered Smart Agriculture Platform | *FastAPI, XGBoost, MobileNetV3, ONNX, React Native*

- Built XGBoost crop and yield models: 93% accuracy over 22 crops, R-squared 0.91 and MAE 1.51 t/ha across 10,429 records.
- Deployed MobileNetV3 ONNX disease and pest models (38 classes, 98.1% unseen accuracy); subject gate cut false accepts 33% to 2%.

SentinelOps AI - Cloud Incident Detection and Recovery | *FastAPI, PostgreSQL, Scikit-learn, Next.js, Docker, Render, Neon*

- Built an 8-cause root-cause engine with read-only falsification tests, naming the correct cause 99% across 20 benchmark seeds.
- Shipped recovery contracts with rehearsal and SLO verification on Render, Vercel and Neon; cut false recovery claims 72% to 0%.

Contexta - Autonomous AI Threat Intelligence Platform | *FastAPI, Next.js, PostgreSQL, Gemini, NetworkX, Redis, Docker*

- Built 5 role-specific Gemini agents reasoning over SHA-256 hashed evidence packs, with deterministic consensus streamed via SSE.
- Engineered a 6-factor BWVS risk score from CVSS, KEV, EPSS and 7-day decay, sealed in an Ed25519 signed hash-chain ledger.

VERITAS - Verified Evolving Reality and Intelligence Tracking | *PyTorch, FastAPI, RAG, BM25, Kafka, SQLite, React*

- Built a bitemporal RAG pipeline with BM25-dense fusion and reranking, scoring 1.00 on TemporalEvidenceBench vs 0.11 basic RAG.
- Trained a 14M-parameter transformer from scratch (RoPE, GQA, SwiGLU, RMSNorm); audit 50/50 correct with 12/12 refusals.

SKILLS

- **Programming Languages:** C, C++, Python, Java, JavaScript, TypeScript, SQL
- **Frontend and Design:** HTML, CSS, React.js, Next.js, Angular.js, Bootstrap, Tailwind CSS, Gradio, Streamlit, Canva, Figma
- **Backend and Database:** Node.js, Express.js, Flask, FastAPI, REST APIs, MySQL, MongoDB, PostgreSQL, Redis, SQLAlchemy
- **Machine Learning and AI:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, OpenCV, YOLO, ONNX, OCR
- **Generative and Agentic AI:** LLMs, Transformers, RAG, LangChain, LangGraph, Prompt Engineering, Multi-Agent Systems
- **DevOps, Cloud and Testing:** Git, GitHub, Docker, Jenkins, Kubernetes, Linux (Ubuntu), CI/CD, AWS (EC2, S3), Render, Pytest
- **CS Fundamentals:** Data Structures and Algorithms, Operating Systems, Computer Networks, DBMS, OOPs, System Design, Agile

COMPETITIVE PROGRAMMING

- **LeetCode** (barath80): Top 10%, 1000+ problems solved, Max Rating 1815, with a 500+ day streak of consistent daily practice
- **CodeChef** (barath80): Max Rating 1183, 222+ problems solved, earning Bronze & Gold Badges through regular contests
- **CodeForces** (sunkireddybarath07): Max Rating 706, an active contest participant building strong problem-solving speed and logic

ACHIEVEMENTS

- **BeachHack National Hackathon'26** - 2nd Runner-Up, Christ University (Built Contexta; Rs. 20,000 Prize) *Feb 2026*
- **QIE International Blockchain Hackathon'26** - Winner (Built NeuroCred, an on-chain credit passport; $2500 USD Prize) *Jan 2026*
- **Hack Karnataka (GDG)'25** - Finalist (Developed AgriSmart AI; Top 2 track ranking among 3,000+ teams nationwide) *Nov 2025*
- **Cardano Hackathon Asia IBW Edition'25** - Winner (Built Lendora AI, a DeFi platform; 5,000+ teams; $2000 Prize) *Nov 2025*